

Name: \_\_\_\_\_



## Decrease by a percentage

### Task A: Decreasing an amount by a percentage

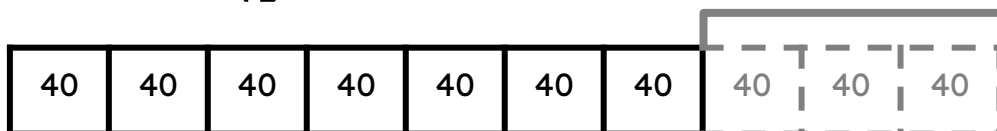
1) Pair up the equivalent questions:

|                     |                    |
|---------------------|--------------------|
| Decrease 50 by 30%  | Work out 80% of 50 |
| Work out 97% of 50  | Increase 50 by 45% |
| Decrease 50 by 20%  | Work out 70% of 50 |
| Work out 145% of 50 | Decrease 50 by 3%  |

2) Fill in the bar model and missing information to show:

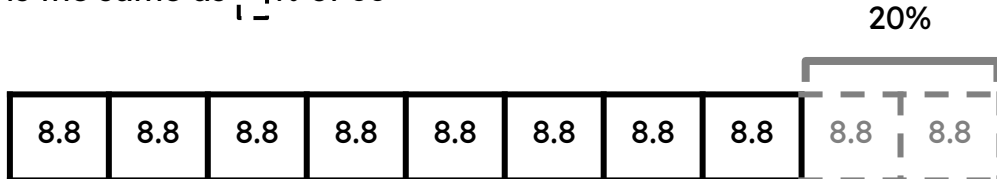
a) 400 decreased by 30% is the same as % of 400

Answer =



b) 88 decreased by 20% is the same as % of 88

Answer =



c) 64 decreased by % is the same as 65% of 64

Answer =



3a) Draw a bar model to work out and represent £124 decreased by 40%

3b) Draw a bar model to work out and represent £72 decreased by 35%

### Task B: Using a multiplier

1) Match the statements with the calculation.

Work out 84% of 400

Decrease 600 by 14%

Work out 86% of 400

Decrease 14 by 16%

Decrease 16 by 61%

Work out 39% of 16

Work out 86% of 600

Decrease 400 by 14%

Work out 84% of 14

Decrease 400 by 16%

$400 \times 0.84$

$16 \times 0.39$

$600 \times 0.86$

$400 \times 0.86$

$14 \times 0.84$



2) Work out the amount of money when £94 has been decreased by 47%

3) Work out the amount of money when £162 has been decreased by 28%

4) A T-shirt was reduced by 24% in a sale.

How much is the T-shirt now?



### Task C: Mixed percentage change

1) Tick the calculations that correctly represent each statement

| Statement           | Calculation      | Tick |
|---------------------|------------------|------|
| Increase £20 by 35% | $20 \times 1.35$ |      |
| Decrease £20 by 35% | $20 \times 0.35$ |      |
| Increase £35 by 20% | $35 \times 0.2$  |      |
| Decrease £35 by 20% | $35 \times 0.8$  |      |
| Increase £45 by 1%  | $45 \times 1.1$  |      |
| Decrease £45 by 1%  | $45 \times 0.99$ |      |



2a) Aisha has £20 to buy a present for her dad. There is a sale on at the shop. Below are some items with their original prices and the discounts to be applied. Which of the items can she afford?

Shorts £30



10% off

T-shirt £25



20% off

Jacket £50



50% off

Hat £22



50% off

Scarf £22



15% off

b) For those she can't afford, what discounts would she need so that she can afford any item in the shop?

3) Laura says,

“If you increase an amount by 13% and then decrease it by 13% you get the same value you started with.”

Explain whether Laura is correct.

